

HU Collection

TMUA Mock Set H Paper 2

Worked Solutions

Kevin Hu 24 September 2026

Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	A	B	D	F	B	C	A	E	D	F
Q	11	12	13	14	15	16	17	18	19	20
Ans	B	A	C	E	D	B	A	C	F	C

Each solution gives the intended route and names the trap the question was built around. Questions are ordered by increasing difficulty.

Question 1. Answer A.

If every term is an integer, then certainly u_1 and u_3 are: necessary. It is not sufficient, because the common difference is $\frac{u_3 - u_1}{2}$, which may be a half-integer: $1, \frac{3}{2}, 2, \frac{5}{2}, \dots$ has u_1 and u_3 integers but u_2 not. The condition “ u_1 and u_2 are integers” would have been necessary and sufficient.

Question 2. Answer B.

A counterexample needs $x^2 > x$ and $x \leq 1$. $x^2 > x$ holds exactly when $x < 0$ or $x > 1$, so the counterexamples are the negative numbers: -1 and $-\frac{1}{2}$. For $0, \frac{1}{2}$ and 1 the hypothesis fails, so they are not counterexamples, and 2 satisfies the conclusion. Two in all.

Question 3. Answer D.

The setup is sound: $64q^2 + 16qr$ is divisible by 8 , so $8 \mid n^2$ exactly when $8 \mid r^2$. But the student misreads the list: $r = 4$ gives $r^2 = 16$, which is divisible by 8 . So the argument fails, and the claim is false: $n = 4$ has $n^2 = 16$ divisible by 8 while $8 \nmid 4$. What is true is that $8 \mid n^2$ forces $4 \mid n$.

Question 4. Answer F.

Split by sign, since multiplying by x reverses inequalities when $x < 0$. For $x > 0$: $x < \frac{1}{x}$ gives $x < 1$, while $\frac{1}{x} < x^2$ gives $x^3 > 1$, that is $x > 1$; these are incompatible. For $x < 0$: $\frac{1}{x} < 0 < x^2$ holds automatically, and $x < \frac{1}{x}$, multiplied by the negative x , gives $x^2 > 1$, so $x < -1$. The solution set is $x < -1$.

Question 5. Answer B.

$x|x|$ is increasing on the whole line, so for $a \geq 0$, f is a sum of increasing functions and is one-to-one. For $a < 0$, $f(0) = 0$ and $f(-a) = a^2 + a(-a) = 0$ with $-a \neq 0$, so f is not one-to-one. So f is one-to-one exactly when $a \geq 0$, and $a > 0$ is sufficient but not necessary: $a = 0$ gives the one-to-one $f(x) = x|x|$.

Question 6. Answer C.

Go through the regions cut by -3 , 2 , 3 and 5 . For $x < -3$ only Carol and Dan are right: two. For $-3 \leq x \leq 2$ only Carol: one. For $2 < x \leq 3$ Alice and Carol: two. For $3 < x < 5$ Alice, Bob, Carol and Dan: four. For $x \geq 5$ Alice, Bob and Dan: three. So the set is $x < -3$ or $2 < x \leq 3$. The endpoint $x = 3$ is included because $9 > 9$ is false, and $x = -3$ is excluded for the same reason.

Question 7. Answer A.

Each statement forces the person on the left to be the opposite type from the speaker, so types alternate around the table. Alternation round a cycle is possible exactly when n is even: I and II. Then exactly half are liars: III. IV fails: for even n there are two alternating patterns, depending on whether a chosen person is a truth-teller or a liar.

Question 8. Answer E.

I: if $b = ar$ and $c = ar^2$ then $b^2 = a^2r^2 = ac$. II fails: $a = b = 0$, $c = 5$ has $b^2 = ac$ but is not a geometric progression. III fails: $a = 1$, $b = 0$, $c = 0$ has $b^2 = ac$ and $a \neq 0$, but the ratio would have to be 0, which is excluded. IV holds: $b \neq 0$ forces a and c non-zero, and then $\frac{b}{a} = \frac{c}{b}$ is a common non-zero ratio. So I and IV.

Question 9. Answer D.

I: take $b = 1 - a$, so $a + b = 1$ and $ab = a(1 - a)$, a product of consecutive integers, which is even. II fails: whatever b is, $a = b + 1$ makes $a + b$ odd, but $a = b$ makes it even. III: $a^2 + b^2 - 2ab = (a - b)^2 = 1$; take $b = a + 1$. IV: $a = 2$. V: if $a + b$ is odd, one of a , b is even. Four are true. II and I differ only in the order of the quantifiers.

Question 10. Answer F.

After a additions and b subtractions the position is $3a - 5b$, in any order. I: $1 = 3 \cdot 2 - 5 \cdot 1$, and repeating that combination reaches every integer. II: $3a - 5b = 1$ needs $a \equiv 2 \pmod{5}$, so $a \geq 2$ and then $b \geq 1$; the minimum is 3 moves. III: $3a - 5b = 2$ needs $a \equiv 4 \pmod{5}$, so $a \geq 4$ and $b \geq 2$, at least 6 moves. IV: $3a = 5b$ with $a + b = 10$ would need $8a = 50$, impossible. So I, II and IV.

Question 11. Answer B.

If x and y have the same sign, or one is zero, then $|x + y| = |x| + |y|$ and the equation fails. So $xy < 0$ throughout: III. With opposite signs, $|x| + |y| - |x + y| = 2 \min(|x|, |y|)$, so S is the set where $\min(|x|, |y|) = \frac{1}{2}$ with opposite signs: two L-shaped paths, each made of two rays, with corners at $(\frac{1}{2}, -\frac{1}{2})$ and $(-\frac{1}{2}, \frac{1}{2})$. So I and II hold. IV fails: on S , $|x| + |y| \geq 2 \cdot \frac{1}{2} = 1$.

Question 12. Answer A.

I is complete: a starting value of 0, and differences that are all multiples of 5. II is complete: the five residue classes cover every integer, and $2^5 - 2 = 30$, $3^5 - 3 = 240$ and $4^5 - 4 = 1020$ are all multiples of 5. III is invalid: $n^2 + 1$ is not an integer adjacent to $n + 1$, so the factors are not five consecutive integers. (The factorisation can be rescued by writing $n^2 + 1 = (n^2 - 4) + 5$, but that is a different argument.)

Question 13. Answer C.

Four values determine a unique polynomial of degree at most 3; here it is $\frac{x^3 + 3x^2 + 2x}{6}$, of degree exactly 3: I. II fails because a quadratic through the first three points would be x^2 , and $3^2 \neq 10$.

III: add any multiple of $x(x-1)(x-2)(x-3)$. IV fails for a subtler reason: for integer coefficients, $a-b$ always divides $p(a) - p(b)$, since $a^k - b^k$ is divisible by $a-b$. But $3-0=3$ does not divide $10-0=10$.

Question 14. Answer E.

I: a non-zero x fails once $\frac{1}{n} < |x|$. II: some n works exactly when $n=1$ works, that is $|x| < 1$. III fails: $C = \{x : x < 1\}$, since the condition for $n=1$ is the strongest. IV fails: $D = \{x : x > 1\}$, since the condition for $n=1$ is the weakest. So I and II. “For every” makes the strongest instance decisive, and “for some” the weakest.

Question 15. Answer D.

I: $0 \leq f \leq 1$ gives $f^2 \leq f$. II: $\int (f - \frac{1}{2})^2 \geq 0$ expands to $\int f^2 - \int f + \frac{1}{4} \geq 0$, so $\int f^2 \geq \frac{1}{4}$. III: $\int xf$ is largest when the available “mass” $\frac{1}{2}$ sits as far right as possible, and it is at most the value for $f=1$ on $[\frac{1}{2}, 1]$, which is $\frac{3}{8}$. Formally, $\int (x - \frac{1}{2})(f(x) - \mathbf{1}_{[1/2, 1]}) dx \leq 0$, because the two factors never share a sign. IV fails: $f(x) = x$ gives $\int xf = \frac{1}{3}$.

Question 16. Answer B.

Use the Euclidean algorithm. d divides the difference $(n+1)^2 + 1 - (n^2 + 1) = 2n + 1$, and hence divides $4(n^2 + 1) - (2n + 1)(2n - 1) = 5$. So $d \in \{1, 5\}$: I holds and IV fails. II: 5 and 10 have highest common factor 5. III: $d(n) = 5$ whenever $5 \mid n^2 + 1$ and $5 \mid 2n + 1$, which happens for every $n \equiv 2 \pmod{5}$.

Question 17. Answer A.

$g(x+1) = f(x+1) - (x+1) = g(x)$, so g is continuous with period 1. I follows (apply the relation n times, forwards or backwards). II: g is bounded, being continuous on $[0, 1]$ and repeating. III fails: $f(x) = x + \sin(2\pi x)$ satisfies the condition but decreases where $1 + 2\pi \cos(2\pi x) < 0$. IV fails: $f(x) = x - 5\sin^2(\pi x)$ has $f(0) = 0$ and $f(\frac{1}{2}) = \frac{1}{2} - 5 < -\frac{1}{2}$. So I and II.

Question 18. Answer C.

I: $(|\sin x| + |\cos x|)^2 = 1 + 2|\sin x \cos x| \geq 1$. III: $(\sin x - \cos x)^2 = 1 - 2\sin x \cos x \leq 2$. II fails whenever $\tan x < 0$. IV: cosine is even, so $\cos(\sin x) = \cos|\sin x| = \sin(\frac{\pi}{2} - |\sin x|)$, and $\frac{\pi}{2} - |\sin x|$ lies in $[\frac{\pi}{2} - 1, \frac{\pi}{2}]$. Also $\cos x$ lies in $[-1, 1]$. Both are inside $[-\frac{\pi}{2}, \frac{\pi}{2}]$, where sine is increasing, so it suffices that $\frac{\pi}{2} - |\sin x| > \cos x$, that is $|\sin x| + \cos x < \frac{\pi}{2}$. This is true because $|\sin x| + \cos x \leq \sqrt{2} < \frac{\pi}{2}$. So I, III and IV.

Question 19. Answer F.

I: $\frac{x+y+z}{3} \geq \sqrt[3]{xyz} = 1$. II: apply the same to xy, yz, zx , whose product is 1. III: $3(x^2 + y^2 + z^2) \geq (x + y + z)^2$, and $(x + y + z)^2 \geq 3(x + y + z)$ by I; divide by 3. IV fails: $x = 4, y = z = \frac{1}{2}$ gives $\frac{1}{4} + 2 + 2 = 4.25 < 5$. In fact IV says $xy + yz + zx \geq x + y + z$, which is not symmetric enough to be forced.

Question 20. Answer C.

$x = 0$ is a double root and $x = 2$ a root, so $f(x) = x^2(x-2)(ax-b)$ with $a > 0$, and $f(1) = -1$ gives $b = a - 1$. So $f(x) = x^2(x-2)(ax - (a-1))$, a one-parameter family. III: $x = 0$ is stationary, and f has a minimum somewhere in $(0, 2)$ since $f(1) < 0 = f(0) = f(2)$. The rest can fail. For

$a = 2$ the fourth root is $\frac{1}{2}$, so $f > 0$ on $(0, \frac{1}{2})$, breaking II, and 0 is itself a local minimum, giving two minima and breaking I. For $a = \frac{1}{2}$, $f(-1) = 0$, breaking IV. Only III is forced.